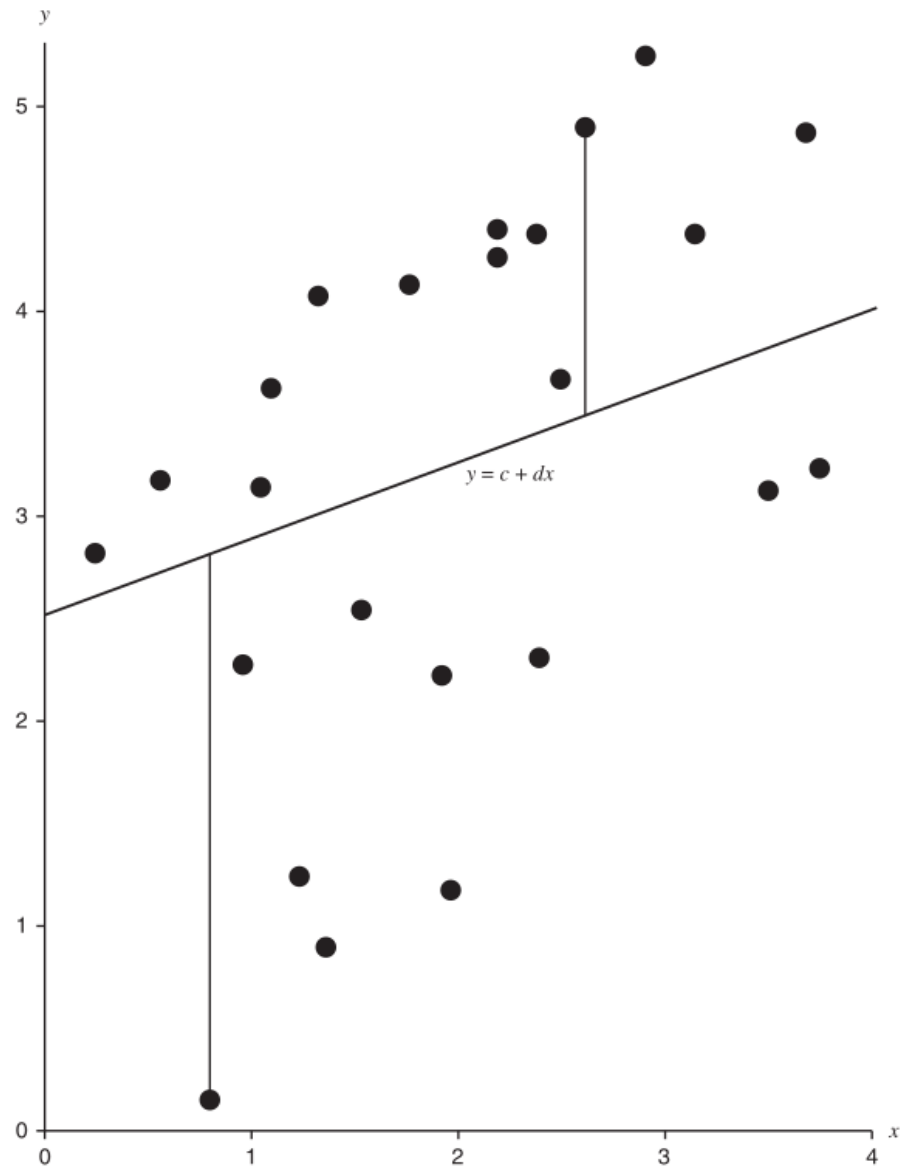


Regresión lineal simple

Hasta ahora: modelamos una variable aleatoria por medio de una distribución que dependía de uno o (más) parámetros.

Ahora: le agregamos que dicha variable aleatoria podría depender del valor de otra variable.

Regresión: estimar la relación entre dos variables



x	y	x	y	x	y	x	y
3.74	3.22	0.20	2.81	1.22	1.23	1.76	4.12
3.66	4.87	2.50	3.71	1.00	3.13	0.51	3.16
0.78	0.12	3.50	3.11	1.29	4.05	2.17	4.40
2.40	2.31	1.35	0.90	0.95	2.28	1.99	1.18
2.18	4.25	2.36	4.39	1.05	3.60	1.53	2.54
1.93	2.24	3.13	4.36	2.92	5.39	2.60	4.89
$\bar{x} = 1.95$	$\bar{y} = 3.18$	$S_{xx} = 22.82$		$S_{yy} = 43.62$		$S_{xy} = 15.48$	

$$Y_i = \alpha + \beta x_i + \varepsilon_i$$

Diagram illustrating the components of the linear regression equation $Y_i = \alpha + \beta x_i + \varepsilon_i$:

- Y_i : variable aleatoria
- α : parámetros de la relación
- β : parámetros de la relación
- x_i : variable "fija"
- ε_i : variable aleatoria

$$Y_i = \alpha + \beta x_i + \varepsilon_i,$$

Y_i : variable de respuesta (“response variable”)

Supongamos que

$$E\varepsilon_i = 0$$

x_i : covariable, variable predictiva
 (“covariate, predictor variable, feature”)

entonces tenemos la función de regresión poblacional:

$$EY_i = \alpha + \beta x_i.$$

(...evitamos decir variable independiente y variable dependiente)

Para enfatizar el hecho de que nuestra inferencia sobre la relación entre Y_i y x_i asume conocimiento de las x_i , podemos escribir:

$$E(Y_i | x_i) = \alpha + \beta x_i$$

Ejemplos de regresión lineal:

$$E(Y_i|x_i) = \alpha + \beta x_i^2$$

$$E(\log Y_i|x_i) = \alpha + \beta (1/x_i)$$

**LINEAL EN LOS
PARAMETROS!!**

Esto NO ES regresión lineal:

$$E(Y_i|x_i) = \alpha + \beta^2 x_i$$

Based on the data $(x_1, y_1), \dots, (x_n, y_n)$, define the following quantities. The *sample means* are

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i. \quad (11.3.5)$$

The *sums of squares* are

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 \quad \text{and} \quad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 \quad (11.3.6)$$

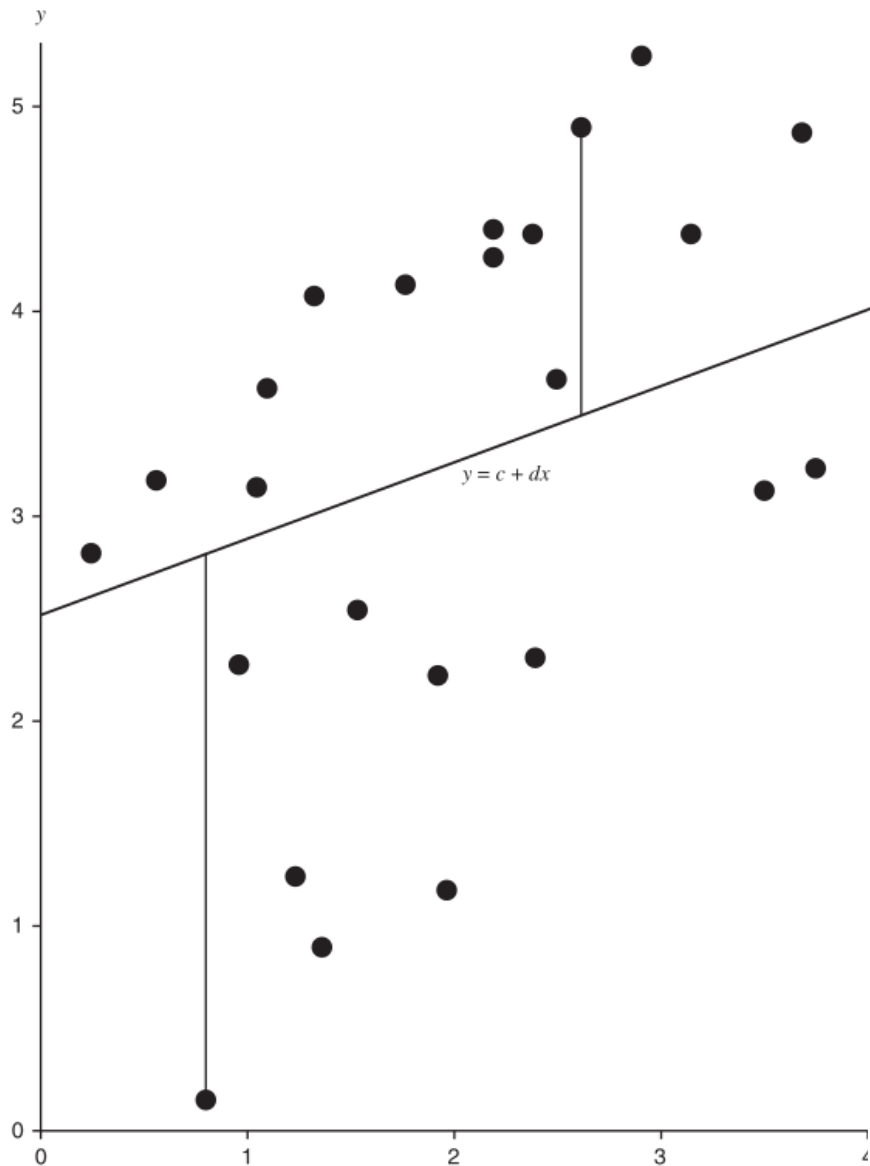
and the *sum of cross-products* is

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}). \quad (11.3.7)$$

Estimadores más comunes para α y β :

$$b = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad a = \bar{y} - b\bar{x}.$$

¿De dónde salen?



Supongamos que la recta tiene la siguiente expresión:

$$y = c + dx,$$

Definimos la Suma de los Residuos Cuadrados:

$$\text{RSS} = \sum_{i=1}^n (y_i - (c + dx_i))^2$$

(residuo: distancia vertical entre cada punto y la recta)

Buscamos a y b tales que RSS sea mínimo

$$\min_{c,d} \sum_{i=1}^n (y_i - (c + dx_i))^2 = \sum_{i=1}^n (y_i - (a + bx_i))^2$$

Reescribimos:

$$\sum_{i=1}^n (y_i - (c + dx_i))^2 = \sum_{i=1}^n ((y_i - dx_i) - c)^2$$

Y utilizamos el teorema 5.2.4, el valor que minimiza esa suma de binomios cuadrados es la media:

$$c = \frac{1}{n} \sum_{i=1}^n (y_i - dx_i) = \bar{y} - d\bar{x}.$$

Reemplazo en RSS y me queda en función de d:

$$\sum_{i=1}^n ((y_i - dx_i) - (\bar{y} - d\bar{x}))^2 = \sum_{i=1}^n ((y_i - \bar{y}) - d(x_i - \bar{x}))^2 = S_{yy} - 2dS_{xy} + d^2S_{xx}.$$

Pido derivada igual a cero:

$$d = \frac{S_{xy}}{S_{xx}}.$$

Si estimo

$$Y = a + b x$$

Es lo mismo que decir

$$X = -(a/b) + (1/b) Y ?$$

En realidad no, porque minimizar el RSS “invertido” da:

$$b' = \frac{S_{xy}}{S_{yy}} \quad \text{and} \quad a' = \bar{x} - b' \bar{y}. \quad (\text{comparar con el original})$$

$$b = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad a = \bar{y} - b \bar{x}.$$

Y por lo tanto la relación quedaría diferente:

$$\hat{y} = -(a'/b') + (1/b')x.$$

Modelos estadísticos para la regresión

Hasta acá no consideramos mucha estadística: los estimadores de α y β salen de una minimización matemática (cuadrados mínimos).
Intervalos de confianza para los estimadores???

Modelo estadístico: modelo condicional normal

$$Y_i \sim n(\alpha + \beta x_i, \sigma^2), \quad i = 1, \dots, n$$

Podríamos escribir:

$$Y_i = \alpha + \beta x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

Con:

$$\varepsilon_1, \dots, \varepsilon_n \text{ are iid } n(0, \sigma^2)$$

Y_i son normales e independientes (aunque no idénticamente distribuidas: igual varianza pero diferente media)

La distribución conjunta de las Y_i es:

$$\begin{aligned} f(\mathbf{y}|\alpha, \beta, \sigma^2) &= f(y_1, \dots, y_n|\alpha, \beta, \sigma^2) \\ &= \prod_{i=1}^n f(y_i|\alpha, \beta, \sigma^2) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-(y_i - (\alpha + \beta x_i))^2 / (2\sigma^2) \right] \\ &= \frac{1}{(2\pi)^{n/2} \sigma^n} \exp \left[- \left(\sum_{i=1}^n (y_i - \alpha - \beta x_i)^2 \right) / (2\sigma^2) \right] \end{aligned}$$

Calculemos los estimadores de máxima verosimilitud para α , β y σ^2

First, we find the maximum likelihood estimates of the three parameters, α , β , and σ^2 . Using the joint pdf in (11.3.24), we see that the log likelihood function is

$$\log L(\alpha, \beta, \sigma^2 | \mathbf{x}, \mathbf{y}) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\sum_{i=1}^n (y_i - \alpha - \beta x_i)^2}{2\sigma^2}.$$

For any fixed value of σ^2 , $\log L$ is maximized as a function of α and β by those values, $\hat{\alpha}$ and $\hat{\beta}$, that minimize

$$\sum_{i=1}^n (y_i - \alpha - \beta x_i)^2.$$

But this function is just the RSS from [Section 11.3.1](#)! There we found that the minimizing values are

$$\hat{\beta} = b = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad \hat{\alpha} = a = \bar{y} - b\bar{x} = \bar{y} - \hat{\beta}\bar{x}.$$

Mismo resultado que con cuadrados mínimos!
Pero ahora podemos estimar incertezas y
hacer inferencia 8)

Falta la estimación de σ^2 :

Thus, the least squares estimators of α and β are also the MLEs of α and β . The values $\hat{\alpha}$ and $\hat{\beta}$ are the maximizing values, for any fixed value of σ^2 . Now, substituting in the log likelihood, to find the MLE of σ^2 we need to maximize

$$-\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}x_i)^2}{2\sigma^2}.$$

This maximization is similar to finding the MLE of σ^2 in ordinary normal sampling (Example 7.2.11) and we leave the details to Exercise 11.28. The MLE of σ^2 , under the conditional normal model, is

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}x_i)^2,$$

σ^2 estimada a través de la varianza de los residuos! (pues asumimos que el valor medio de los epsilon es cero)

$\hat{a}$ y $\hat{b}$ son estimadores insesgados, pero $\hat{s}^2$ tiene sesgo:

$$E\hat{\sigma}^2 = \frac{n-2}{n}\sigma^2.$$

Entonces se usa el siguiente estimador corregido (insesgado):

$$S^2 = \frac{n}{n-2}\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}x_i)^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{\varepsilon}_i^2$$

Theorem 11.3.3 Under the conditional normal regression model (11.3.22), the sampling distributions of the estimators $\hat{\alpha}$, $\hat{\beta}$, and S^2 are

$$\hat{\alpha} \sim n\left(\alpha, \frac{\sigma^2}{nS_{xx}} \sum_{i=1}^n x_i^2\right), \quad \hat{\beta} \sim n\left(\beta, \frac{\sigma^2}{S_{xx}}\right),$$

with

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = \frac{-\sigma^2 \bar{x}}{S_{xx}}.$$

(falta un signo igual!)

Furthermore, $(\hat{\alpha}, \hat{\beta})$ and S^2 are independent and

$$\frac{(n-2)S^2}{\sigma^2} \sim \chi_{n-2}^2.$$

Con esto podemos hacer inferencia:

Inferences regarding the two parameters α and β are usually based on the following two Student's t distributions. Their derivations follow immediately from the normal and χ^2 distributions and the independence in Theorem 11.3.3. We have

$$\frac{\hat{\alpha} - \alpha}{S \sqrt{(\sum_{i=1}^n x_i^2) / (nS_{xx})}} \sim t_{n-2} \quad (11.3.32)$$

and

$$\frac{\hat{\beta} - \beta}{S / \sqrt{S_{xx}}} \sim t_{n-2}. \quad (11.3.33)$$

The joint distribution of these two t statistics is called a *bivariate Student's t distribution*. This distribution is derived in a manner analogous to the univariate case. We use the fact that the joint distribution of $\hat{\alpha}$ and $\hat{\beta}$ is bivariate normal and the same variance estimate S is used in both univariate t statistics. This joint distribution would be used if we wanted to do simultaneous inference regarding α and β . However, we shall deal only with the inferences regarding one parameter at a time.

Contraste de hipótesis para beta

To test

$$H_0: \beta = 0 \quad \text{versus} \quad H_1: \beta \neq 0, \quad (11.3.34)$$

using (11.3.33), we reject H_0 at level α if

$$\left| \frac{\hat{\beta} - 0}{S/\sqrt{S_{xx}}} \right| > t_{n-2, \alpha/2},$$

or, equivalently, if

$$\frac{\hat{\beta}^2}{S^2/S_{xx}} > F_{1, n-2, \alpha}. \quad (11.3.35)$$

(distribución F =
Snedecor = Snedecor-
Fisher
~ cociente de dos chi2

Recalling the formula for $\hat{\beta}$ and that $\text{RSS} = \sum \hat{\epsilon}_i^2$, we have

$$\frac{\hat{\beta}^2}{S^2/S_{xx}} = \frac{S_{xy}^2/S_{xx}}{\text{RSS}/(n-2)} = \frac{\text{Regression Sum of Squares}}{\text{Residual Sum of Squares/df}}.$$

Total Sum of Squares = Regression Sum of Squares + Residual Sum of Squares

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (11.3.36)$$

Típica “tabla ANOVA” como salida de la regresión lineal:

Table 11.3.2 ANOVA table for simple linear regression

Source of variation	Degrees of freedom	Sum of squares	Mean square	<i>F</i> statistic
Regression (slope)	1	Reg. SS= S_{xy}^2/S_{xx}	MS(Reg)= Reg. SS	$F = \frac{\text{MS(Reg)}}{\text{MS(Resid)}}$
Residual	$n - 2$	RSS = $\sum \hat{\epsilon}_i^2$	MS(Resid)= RSS/($n - 2$)	
Total	$n - 1$	SST= $\sum (y_i - \bar{y})^2$		

Ejemplo: un estadístico F muy grande significa que el p es muy chico: pendiente beta significativamente distinta de cero

ANOVA table for grape data

Source of variation	Degrees of freedom	Sum of squares	Mean square	<i>F</i> statistic
Regression	1	6.66	6.66	50.23
Residual	10	1.33	.133	
Total	11	7.99		

Intervalo de confianza para beta

Expression (11.3.33) can be used to construct a $100(1 - \alpha)\%$ confidence interval for β given by

$$\hat{\beta} - t_{n-2, \alpha/2} \frac{S}{\sqrt{S_{xx}}} < \beta < \hat{\beta} + t_{n-2, \alpha/2} \frac{S}{\sqrt{S_{xx}}}. \quad (11.3.37)$$

Also, a level α test of $H_0: \beta = \beta_0$ versus $H_1: \beta \neq \beta_0$ rejects H_0 if

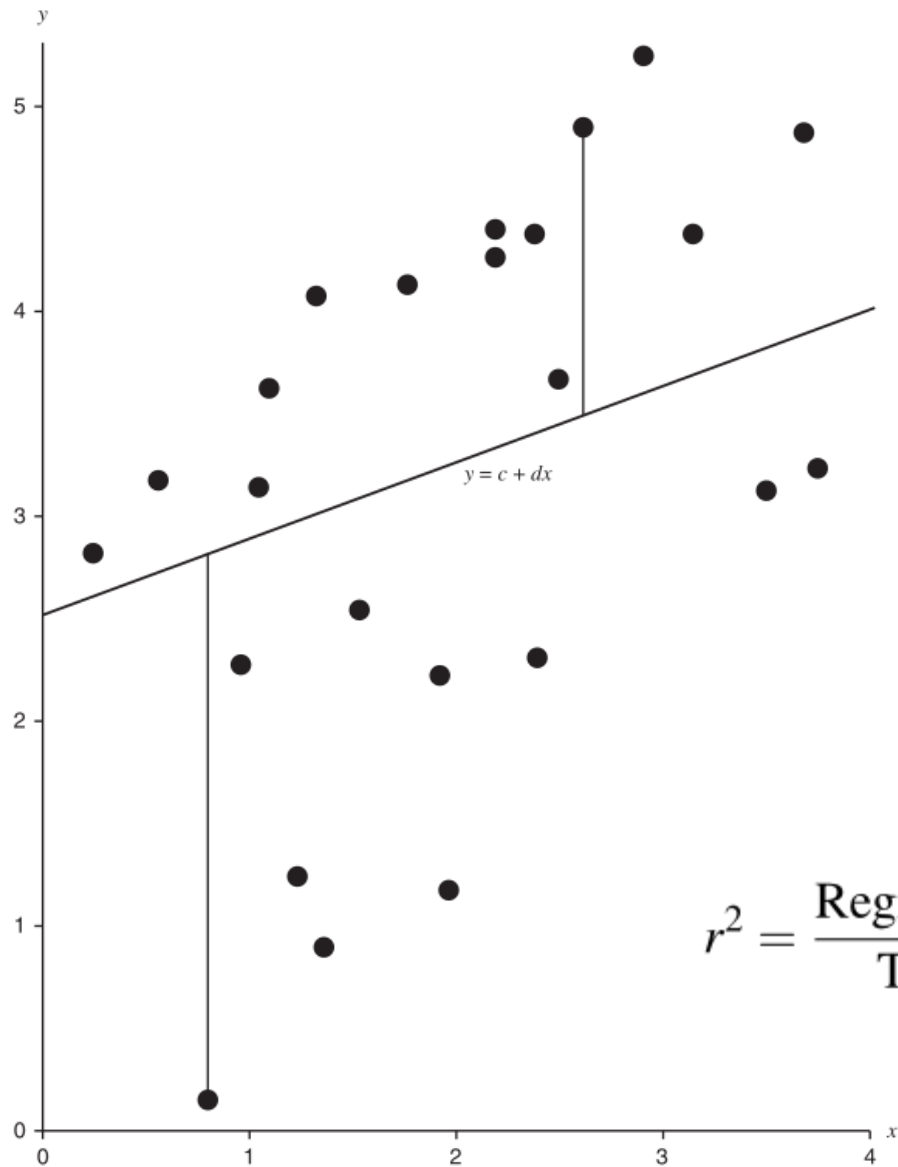
$$\left| \frac{\hat{\beta} - \beta_0}{S/\sqrt{S_{xx}}} \right| > t_{n-2, \alpha/2}. \quad (11.3.38)$$

Bondad del ajuste

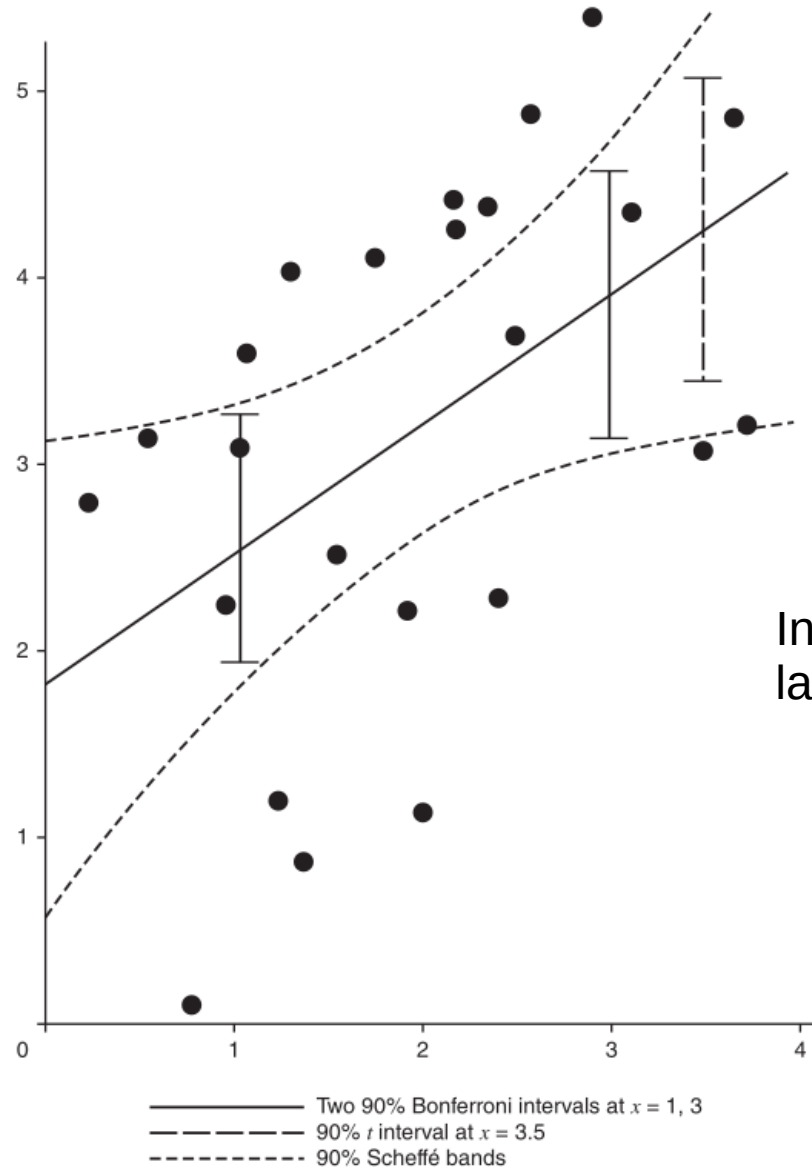
A statistic that is used to quantify how well the fitted line describes the data is the *coefficient of determination*. It is defined as the ratio of the regression sum of squares to the total sum of squares. It is usually referred to as r^2 , and can be written in the various forms

$$r^2 = \frac{\text{Regression Sum of Squares}}{\text{Total Sum of Squares}} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{S_{xy}^2}{S_{xx}S_{yy}}.$$

The coefficient of determination measures the proportion of the total variation in $y_1, \dots, y_n$ (measured by S_{yy}) that is explained by the fitted line (measured by the regression sum of squares). From (11.3.36), $0 \leq r^2 \leq 1$. If $y_1, \dots, y_n$ all fall exactly on the fitted line, then $y_i = \hat{y}_i$ for all i and $r^2 = 1$. If $y_1, \dots, y_n$ are not close to the fitted line, then the residual sum of squares will be large and r^2 will be near 0. The coefficient of determination can also be (perhaps more straightforwardly) derived as the square of the sample correlation coefficient of the n pairs $(y_1, x_1), \dots, (y_n, x_n)$ or of the n pairs $(y_1, \hat{y}_1), \dots, (y_n, \hat{y}_n)$.



$$r^2 = \frac{\text{Regression Sum of Squares}}{\text{Total Sum of Squares}} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{S_{xy}^2}{S_{xx}S_{yy}}.$$



Intervalo de confianza para la regresión!

— Two 90% Bonferroni intervals at $x = 1, 3$
- - - 90% t interval at $x = 3.5$
- - - 90% Scheffé bands

Intervalo de confianza para la regresión

Theorem 11.3.6 *Under the conditional normal regression model (11.3.22), the probability is at least $1 - \alpha$ that*

$$\begin{aligned} \hat{\alpha} + \hat{\beta}x - M_{\alpha}S\sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} \\ < \alpha + \beta x < \hat{\alpha} + \hat{\beta}x + M_{\alpha}S\sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}, \end{aligned} \quad (11.3.43)$$

simultaneously for all x , where $M_{\alpha} = \sqrt{2F_{2,n-2,\alpha}}$.

(sin demostración... usa Bonferroni :)

